Model-Free and Model-Based Reinforcement Learning for Decision Making in Autonomous Ground Vehicles

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***Abstract*—Autonomous Ground Vehicles (AGVs) require decision-making mechanisms capable of operating under uncertainty, nonlinear dynamics, and physical constraints. Reinforcement Learning (RL) provides a general framework for sequential decision making in such systems; however, learning performance depends strongly on whether the agent has access to a model of the environment. Model-free methods offer flexibility and generality but often suffer from poor sample efficiency, while model-based approaches achieve stable and efficient control when accurate models are available.**

This paper presents a comparative study of model-free and model-based reinforcement learning within a unified Markov Decision Process (MDP) framework. After introducing the autonomy stack and situating decision making within it, the mathematical foundations of MDPs and RL are reviewed. The CartPole balancing task is then formulated and solved using a model-free Q-learning agent and a model-based Linear Quadratic Regulator (LQR). The comparison highlights fundamental trade-offs in learning behavior, stability, and sample efficiency, motivating approaches that seek to bridge the gap between purely model-free and purely model-based methods.

***Index Terms*—Reinforcement Learning, Model-Free Control, Model-Based Control, Markov Decision Processes, Autonomous Systems**

I. Introduction

Autonomous Ground Vehicles (AGVs) are driven by the promise of improved safety, efficiency, and reliability in transportation systems. Levels of autonomy range from fully manual operation (Level 0) to full autonomy with human-level perception and decision making (Level 5). Most current AGVs operate at intermediate levels (Levels 2–3), where autonomy is limited to structured environments, predefined routes, and basic obstacle avoidance.

Achieving higher levels of autonomy requires advances across the autonomy stack, encompassing perception, decision making, and control. While perception and low-level control have advanced significantly, decision making under dynamic conditions and uncertainty remains a central challenge. Autonomous agents must select actions sequentially while respecting physical constraints and long-term objectives.

This paper focuses on decision making and examines reinforcement learning as a framework for learning control policies through interaction. In particular, we study how access to environment models fundamentally alters learning behavior by contrasting model-free and model-based approaches within a common mathematical framework.

II. The Autonomy Stack

The autonomy stack enables an autonomous system to perceive its environment, decide on actions, and execute control commands in a closed feedback loop.

A. Perception

Perception estimates the vehicle state and surrounding environment using heterogeneous sensors such as cameras, LiDAR, and radar. Due to sensor imperfections, sensor fusion and probabilistic state estimation are required. Localization is a core outcome of perception and is commonly achieved using SLAM-based techniques.

B. Decision Making

Decision making maps perceived state estimates to actions while encoding trade-offs between safety, performance, and feasibility. Rule-based planners perform well in structured environments but scale poorly as variability increases, motivating learning-based approaches capable of reasoning over continuous state spaces and long horizons.

C. Low-Level Control and Feedback

Low-level control translates decisions into actuator commands while enforcing physical and dynamic limits. Executed actions alter the system state, which is re-observed by perception, closing the perception–decision–control loop.

Decision making lies at the core of this loop and must reason sequentially under uncertainty, motivating a formal treatment using Markov Decision Processes.

III. Markov Decision Processes

Markov Decision Processes (MDPs) provide a mathematical framework for modeling sequential decision-making problems. At each time step, an agent observes a state, selects an action, and receives a reward and a new state.

A. MDP Dynamics

The interaction between the agent and the environment is governed by the transition dynamics

$$p(s', r \mid s, a) = \Pr\{S_{t+1} = s', R_{t+1} = r \mid S_t = s, A_t = a\}. \tag{1}$$

Under the Markov assumption, future system evolution depends only on the current state and action. An MDP is fully defined by the tuple $(\mathcal{S}, \mathcal{A}, P, R, \gamma)$.

B. Reward, Return, and Value Functions

The reward provides an instantaneous measure of performance. Long-term objectives are captured by the return

$$G_t = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}. \tag{2}$$

The state-value and state–action value functions are defined as

$$v_\pi(s) = \mathbb{E}_\pi[G_t \mid S_t = s], \quad q_\pi(s, a) = \mathbb{E}_\pi[G_t \mid S_t = s, A_t = a]. \tag{3}$$

IV. Model-Free and Model-Based Reinforcement Learning

A. Model-Free Learning

In model-free reinforcement learning, the agent does not have access to the transition dynamics and must learn solely from interaction. Q-learning estimates the optimal state–action value function using temporal-difference updates:

$$Q_{t+1}(s, a) = Q_t(s, a) + \alpha \left(r_{t+1} + \gamma \max_{a'} Q_t(s_{t+1}, a') - Q_t(s, a) \right). \tag{4}$$

Although model-free methods are broadly applicable, they typically require extensive sampling to converge and may exhibit unstable behavior during early learning.

B. Model-Based Control

Model-based approaches assume access to system dynamics, enabling planning and control without extensive interaction. When accurate models are available, these methods achieve stable and sample-efficient control, but their applicability is limited when modeling is difficult or inaccurate.

V. CartPole Benchmark

The CartPole system provides a controlled benchmark for comparing model-free and model-based approaches on the same physical system. The task consists of stabilizing an inverted pendulum mounted on a cart by applying horizontal forces.

A. Model-Free Formulation

The state at time t is defined as

$$s_t = (x_t, \dot{x}_t, \theta_t, \dot{\theta}_t), \tag{5}$$

with learning performed using Q-learning and an ϵ-greedy exploration strategy.

B. Model-Based Formulation

The dynamics are linearized about the upright equilibrium,

$$\dot{x} = Ax + Bu, \tag{6}$$

and controlled using a Linear Quadratic Regulator (LQR).

C. Comparison

Model-free learning improves performance gradually through interaction but requires extensive sampling. In contrast, model-based control achieves immediate stability once the system model is available.

VI. Conclusion and Outlook

This paper compared model-free and model-based reinforcement learning approaches for decision making in autonomous systems using the CartPole problem as a benchmark. Model-free methods offer flexibility but suffer from high sample complexity, while model-based methods provide efficient and stable control when accurate models are available.

In practice, accurate models are often difficult to obtain, while purely model-free approaches may be impractical due to sampling constraints. This motivates intermediate approaches that incorporate partial system knowledge without requiring full models. In subsequent work, this idea is explored through physics-informed reinforcement learning applied to a local path-following problem using a vehicle bicycle model.

Figures

Table 1. Comparison between different automation levels.

Organisation	Level 0	Level 1	Level 2	Level 3	Level 4	Level 5
BASt, SAE, OICA	Driver Only	Driver Assisted	Partial Automation	Conditional Automation	High Automation	Full Automation
NHSTA	No	Function Specific	Combined Function	Limited Self-Driving	Fully Self-Driving	

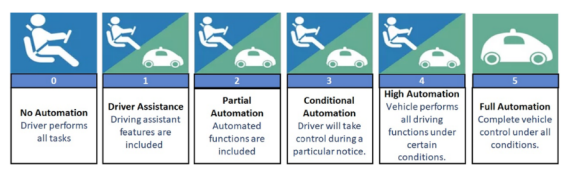


Fig. 1. Different levels of AGV autonomy.

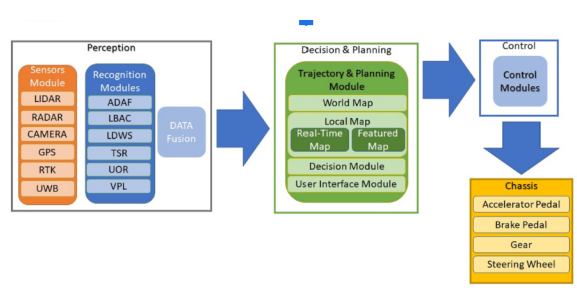


Fig. 2. Components of the autonomy stack in AGVs.

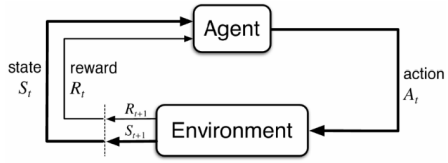


Fig. 3. Markov Decision Process interaction loop.

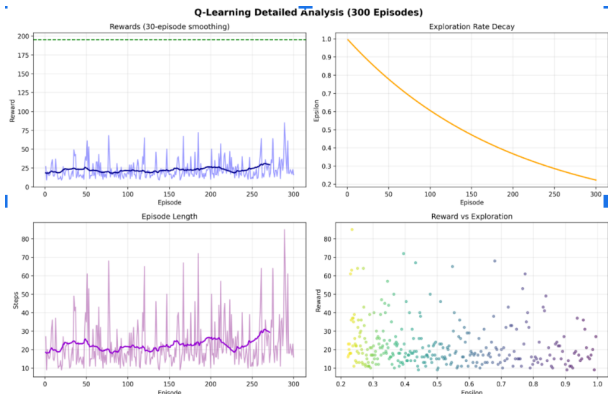


Fig. 4. Model-free Q-learning training behavior.

References

[1] X. Wang *et al.*, "Sensors and sensor fusion for autonomous vehicles: A review," *Sensors*, vol. 21, no. 706, 2021.

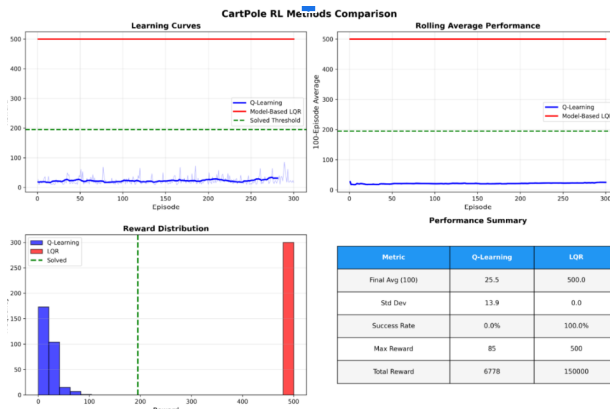
[2] L. P. Kaelbling, M. L. Littman, and A. W. Moore, "Reinforcement learning: A survey," *Journal of Artificial Intelligence Research*, vol. 4, pp. 237–285, 1996.

[3] D. W. Sari, S. Dwijayanti, and B. Y. Suprapto, "Development of autonomous vehicle navigation in unstructured environments," *Eastern-European Journal of Enterprise Technologies*, 2025.

[4] M. Ghasemi and D. Ebrahimi, "Introduction to reinforcement learning," arXiv:2408.07712, 2024.

[5] R. S. Sutton and A. G. Barto, *Reinforcement Learning: An Introduction*, 2nd ed. MIT Press, 2018.

[6] A. Assaad, "RL-CartPole: Model-Free and Model-Based Control Comparison," GitHub repository, 2025. [Online]. Available: https://github.com/ahmadmohannad/RL-Cartpole



Metric	Q-Learning	LQR
Final Avg (100)	25.5	500.0
Std Dev	13.9	0.0
Success Rate	0.0%	100.0%
Max Reward	85	500
Total Reward	6778	150000

Fig. 5. Comparison between model-free Q-learning and model-based LQR control.